CS342 · VISUAL PROGRAMMING

Midterm 462 Solutions

Length Converter

Muharib

01 The analysis

In the picture	In code
4 white boxes	**tCm, tM, tDm, tC** (JTextField)
4 texts	"length in cm" · "= m:" · "dm :" · "cm :" (JLabel)
2 buttons on their own row	GridLayout(2, 1): one row for the fields, one for the buttons
A negative number	A custom NegativeValueException
Not a number	NumberFormatException from parseInt itself

Formula	With 253
m = cm / 100	253 ÷ 100 = 2
dm = (cm % 100) / 10	Remainder 53 → 53 ÷ 10 = 5
c = cm % 10	Remainder = 3

02 The full solution

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
public class Converter extends JFrame {
    private JTextField tCm, tM, tDm, tC;
    private JLabel l1, l2, l3, l4;
    private JButton bConvert, bCancel;
    public Converter() {
        setTitle("Converter");
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        Container c = getContentPane();
        c.setLayout(new GridLayout(2, 1));
        JPanel p1 = new JPanel(new FlowLayout());     // row 1: fields
        JPanel p2 = new JPanel(new FlowLayout());     // row 2: buttons
        l1  = new JLabel("length in cm");
        tCm = new JTextField(10);
        l2  = new JLabel("= m:");
        tM  = new JTextField(3);
        l3  = new JLabel("dm :");
        tDm = new JTextField(3);
        l4  = new JLabel("cm :");
        tC  = new JTextField(3);
        bConvert = new JButton("Convert");
        bCancel  = new JButton("Cancel");
        p1.add(l1); p1.add(tCm);
        p1.add(l2); p1.add(tM);
        p1.add(l3); p1.add(tDm);
        p1.add(l4); p1.add(tC);
        p2.add(bConvert); p2.add(bCancel);
        c.add(p1);
        c.add(p2);
```

```
        pack();
        bConvert.addActionListener(new BConvert());
        bCancel.addActionListener(new BCancel());
    }
```

```
    class BConvert implements ActionListener {
        public void actionPerformed(ActionEvent e) {
            try {
                int cm = Integer.parseInt(tCm.getText());
                if (cm < 0) throw new NegativeValueException();
                int m  = cm / 100;
                int dm = (cm % 100) / 10;
                int c  = cm % 10;
                tM.setText("" + m);
                tDm.setText("" + dm);
                tC.setText("" + c);
            }
            catch (NegativeValueException ex) {
                JOptionPane.showMessageDialog(null, "enter a positive value",
                        "Input Error", JOptionPane.ERROR_MESSAGE);
            }
            catch (NumberFormatException ex) {
                JOptionPane.showMessageDialog(null, "Enter the seconds as a numerical
Value",
                        "Error: Input Format", JOptionPane.ERROR_MESSAGE);
            }
        }
    }
    class BCancel implements ActionListener {
        public void actionPerformed(ActionEvent e) {
            tCm.setText("");
            tM.setText("");
            tDm.setText("");
            tC.setText("");
        }
    }
    public static void main(String[] args) {
        Converter f = new Converter();
        f.setVisible(true);
```

```
    }
}
class NegativeValueException extends Exception { }
```

Exactly the 461 template. The only differences: one custom exception (negative), the formula, and the row layout with GridLayout.

The number message in the question reads "Enter the seconds as a numerical Value", so the solution copies it as is.



